# TMBD 5000 Movies

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# Introduction

## Background

- Movies are one of the most influential media sources that impact modern day culture
- Movie popularity highlights what consumers value in a movie and what is not valued
- There are a vast variety of different movies that have different genres and languages
- Understanding what drives success of a movie is extremely important for producers and investors when making decisions about movie creation
- Movie success is critical for not only financial success but cultural impact on society as well

# Introduction

## What are we aiming to achieve?

- Explore movie data and predict revenue characteristics based on outside variables using Linear Regression
- Explore Popularity based on interactive variables by building Classification trees
- Use predictors to determine end revenue and popularity of a movie

# Dataset Description

- Obtained from Kaggle
- Contains observations from over 5,000 movies
- This dataset was generated from The Movie Database API.
- API also provides access to data on many additional movies, actors and actresses, crew members, and TV shows.
- Includes both numerical and Categorical data making it easily accessible for regression and classification analysis

# Variables of Interest

Target Variables:

- Revenue (Quantitative)
- Popularity (Qualitative)

# Explanatory and Grouped Variables

## Variables Used for Revenue

| Variable Name | Variable Type | Variable Description |
| --- | --- | --- |
| Budget | Quantitative | Movie Budget in US Dollars |
| Vote Count | Quantitative | Number of Votes per movie based on audiences |
| Popularity | Quantitative | Overall popularity score of movies |
| Runtime | Quantitative | How long each movie is in minutes |
| Vote Average | Quantitative | Movie rating scale from 0-10, 0 being the worst, 10 being the best |
| Season | Categorical | The seasons for movie releases, categorized as Winter, Spring, Summer, Fall |
| Original Language | Categorical | Original language a movie was created in, including en, es, fr, etc |
| Budget * Vote Count | Quantitative Interaction | Interaction between Budget and Vote Count to see the effects these two variables have together |

# Research Question for Revenue variable

**What variables can help determine what a movies revenue outcome will be and can they help predict new movie outcomes**

- **H0**: There is no relationship between the predictors and revenue.
- **H1**: At least one predictor is associated with revenue.

# What Parameters Were Tuned?

## Revenue Parameters

- **Number of predictors** - These were selected using the lowest validation error

- **Log Transformation** - Was used to reduce skewness, stabilize the variance and improve overall model assumptions

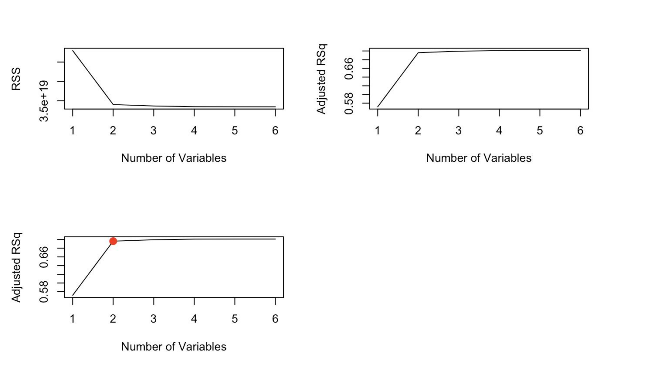
- **Train/Test Split 80/20** - Standard balance between training size and the evaluation reliability

# Explanatory and Grouped Variables

## How were the interactions decided for revenue?

- Through use of Subset Selection, we found that for predicting revenue, two predictors was the best selection.

- It was determined that Budget and Vote Count were the most influential predictors.




| | budget | popularity | runtime | vote_average | vote_count | release_date |
|---|---|---|---|---|---|---|
| 1 ( 1 ) | " " | " " | " " | " " | "*" | " " |
| 2 ( 1 ) | "*" | " " | " " | " " | "*" | " " |
| 3 ( 1 ) | "*" | " " | " " | " " | "*" | "*" |
| 4 ( 1 ) | "*" | "*" | " " | " " | "*" | "*" |
| 5 ( 1 ) | "*" | "*" | " " | "*" | "*" | "*" |
| 6 ( 1 ) | "*" | "*" | "*" | "*" | "*" | "*" |

*Figures 1 and 2 - Model Selection*

# Linear Regression Model

## Did the interaction show a strong relationship to revenue?

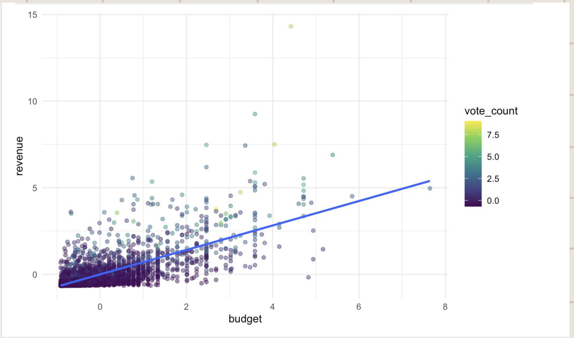
- Train/Test Performance - Since the train RMSE was 0.529 and the test RMSE was 0.545 it showed that there was no strong overfitting indicated and that the model generalized reasonably well overall.

- This determined that there is a strong relationship between the interaction of budget and vote count to revenue.

```
Call:
lm(formula = revenue ~ budget * vote_count + popularity + runtime +
    vote_average, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-3.9012 -0.1985 -0.0620  0.1020  6.5100

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)       -0.040470   0.011047  -3.663 0.000254 ***
budget             0.353115   0.014156  24.945  < 2e-16 ***
vote_count         0.389014   0.019220  20.240  < 2e-16 ***
popularity         0.069782   0.014602   4.779 1.86e-06 ***
runtime           -0.005780   0.011571  -0.500 0.617459
vote_average       0.016702   0.012795   1.305 0.191904
budget:vote_count  0.090676   0.006732  13.469  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.53 on 2576 degrees of freedom
Multiple R-squared:  0.7199,    Adjusted R-squared:  0.7193
F-statistic:  1104 on 6 and 2576 DF,  p-value: < 2.2e-16

    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
-0.74226 -0.45576 -0.24466  0.02642  0.08718  6.76827
[1] 0.5292913
[1] 0.5446904
```

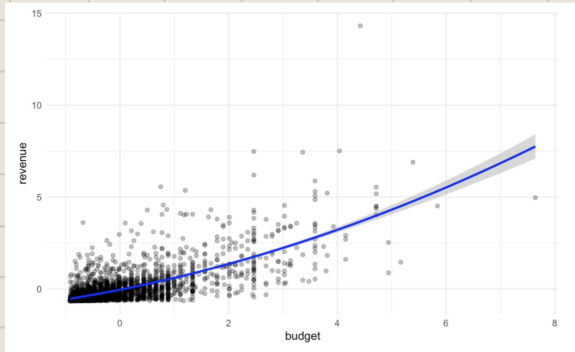
# Linear Regression Model

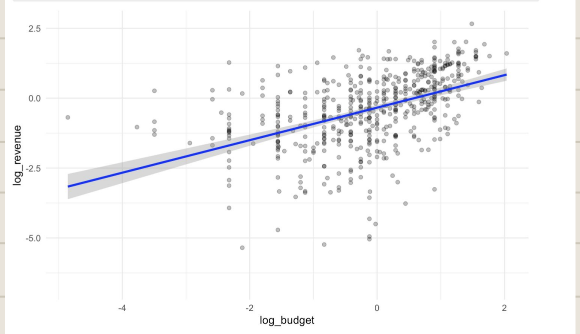
## Did the interaction show a strong relationship to revenue?




This plot shows some clustering as well as extreme outliers.

This plot suggests there in nonlinearity in this model.




This is the log transformed data plotted, to help stabilize the variance and improve overall model assumptions

# Linear Regression Model

## Did the original language of a movie impact revenue?

- In addition to budget and vote count, we decided to see if there were other factors that we didn't account for that impacted revenue.

- Based on the model,  the original language of movies in this dataset did not play a main role in movie revenue as the p-value was 0.40 which shows that it was not statistically significant.

```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         124545063    3344421  37.240   <2e-16 ***
original_languageaf -114665092  186299564  -0.615   0.5383    
original_languagecn  -61584335   83369381  -0.739   0.4601    
original_languageda  -72222009   93194800  -0.775   0.4384    
original_languagede  -82949519   62179855  -1.334   0.1823    
original_languagees  -90917694   48210732  -1.886   0.0594 .  
original_languagefa -123645063  186299564  -0.664   0.5069    
original_languagefr  -91620408   37403728  -2.449   0.0144 *  
original_languagehe -113419214  186299564  -0.609   0.5427    
original_languagehi -104438432   70482661  -1.482   0.1385    
original_languageid -122270182  131754910  -0.928   0.3535    
original_languageis -124545052  186299564  -0.669   0.5038    
original_languageit -117476244   76117731  -1.543   0.1228    
original_languageja  -43278705   51770016  -0.836   0.4032    
original_languageko  -73521540   83369381  -0.882   0.3779    
original_languagenb -120385385  186299564  -0.646   0.5182    
original_languagenl -111183505  131754910  -0.844   0.3988    
original_languageno -122560401  186299564  -0.658   0.5107    
original_languagepl -113845063  186299564  -0.611   0.5412    
original_languagept -106425824  131754910  -0.808   0.4193    
original_languagero -123359280  186299564  -0.662   0.5079    
original_languageru -107782891   76117731  -1.416   0.1569    
original_languagete  -24545063  186299564  -0.132   0.8952    
original_languageth -106493941  131754910  -0.808   0.4190    
original_languagevi -123906063  186299564  -0.665   0.5060    
original_languagexx  -69284505  186299564  -0.372   0.7100    
original_languagezh  -45008100   51770016  -0.869   0.3847    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 186300000 on 3202 degrees of freedom
Multiple R-squared:  0.008409,	Adjusted R-squared:  0.0003577 
F-statistic: 1.044 on 26 and 3202 DF,  p-value: 0.4019
```

# Linear Regression Model

## Did the season of a movie release impact revenue?

- Unlike original language, our created season variable did hold some significance in months like spring and summer.

- Winter did not hold any statistical significance as the p-value was 0.41.

- Even though there was significance, it was not as impactful as budget and vote count on revenue.

```
Call:
lm(formula = revenue ~ season + budget + popularity + vote_average,
    data = movies)

Residuals:
       Min         1Q     Median         3Q        Max
-934327136  -48702491   -8533488   28429327 1980170901

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.861e+08  1.610e+07 -11.560   <2e-16 ***
seasonSpring  1.376e+07  5.784e+06   2.380   0.0174 *
seasonSummer  1.365e+07  5.538e+06   2.464   0.0138 *
seasonWinter  4.638e+06  5.644e+06   0.822   0.4113
budget        2.371e+00  5.173e-02  45.839   <2e-16 ***
popularity    1.664e+06  6.598e+04  25.224   <2e-16 ***
vote_average  2.457e+07  2.471e+06   9.942   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 115100000 on 3222 degrees of freedom
Multiple R-squared:  0.6187,    Adjusted R-squared:  0.618
F-statistic: 871.3 on 6 and 3222 DF,  p-value: < 2.2e-16
```

# Conclusion

## The significance of variables

- Using Linear Regression, we showed that revenue is greatly affected by budget and audience engagement.

- Seeing as budget and vote count increase, revenue increases, this model can be used to predict movie revenue outcomes based on the overall budget.

- We also showed that original language did not have any significance while some seasons had slight significance even when others had none.

# Explanatory and Grouped Variables

## Variables Used for Popularity

| Variable Name | Variable Type | Description |
|---|---|---|
| Budget | Quantitative | Budget for Movie in USD |
| Revenue | Quantitative | Production salves from a movie in USD |
| Vote_Count | Quantitative | Number of rating votes counted for each movie |
| Runtime | Quantitative | The amount of minutes in a movie |
| Vote_Average | Quantitative | Average of vote ratings |
| Num_Companies | Quantitative | Number of companies that produced a given movie |
| Num_Languages | Quantitative | Number of different language in a movie |
| Num_Genres | Quantitative | Number of different genres in a movie |
| Num_Keywords | Quantitative | Number of different keywords in a movie |
| Movie_Age | Quantitative | Age of the movie derived from release_date |

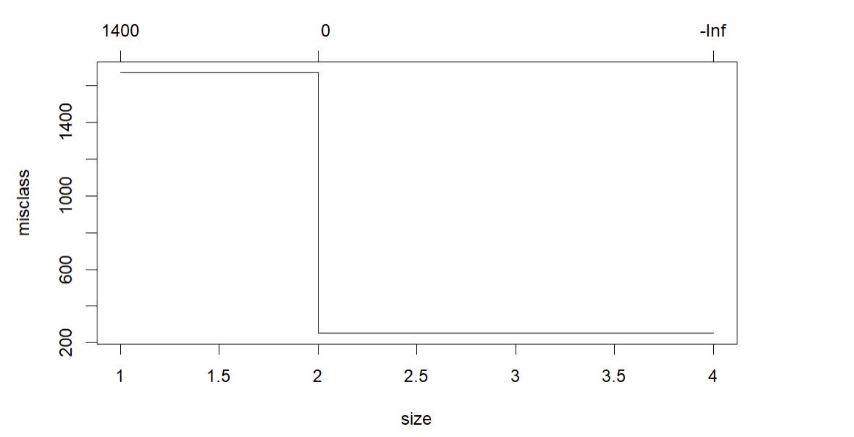
# Research Question to predict Popularity

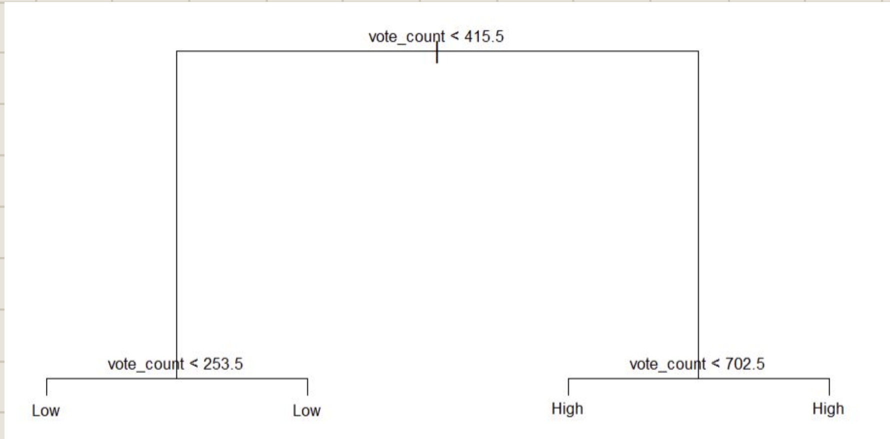
**What specific variables make a movie popular and what are the biggest differences between popular and less popular movies?**

In order to answer this question, I decided that a classification tree method would be best. In order to create this I turned the popularity numerical variable into a categorical one with two groups: Low and High based on if popularity scored above or below the median.

# Tuning Classification Tree

- Initial Classification tree tuned through cv reported that smallest error occurs at 2 predictors
- Tree was very basic, using only one predictor
- Labels observations based on vote_count
- Tree does not help answer Research Question, so other variables will be needed

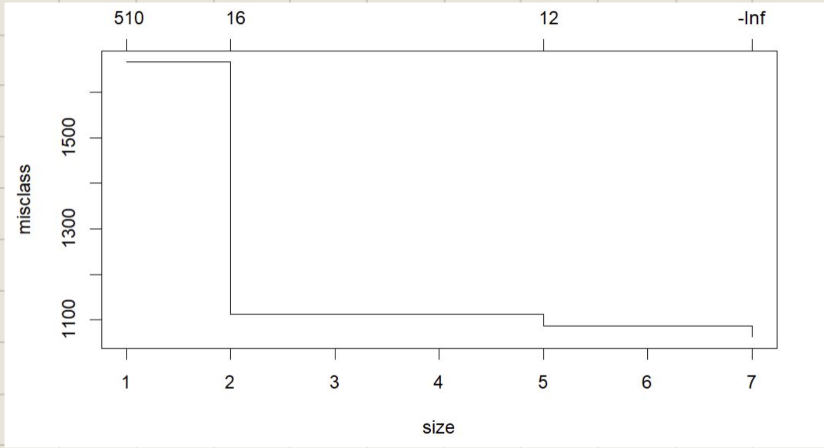


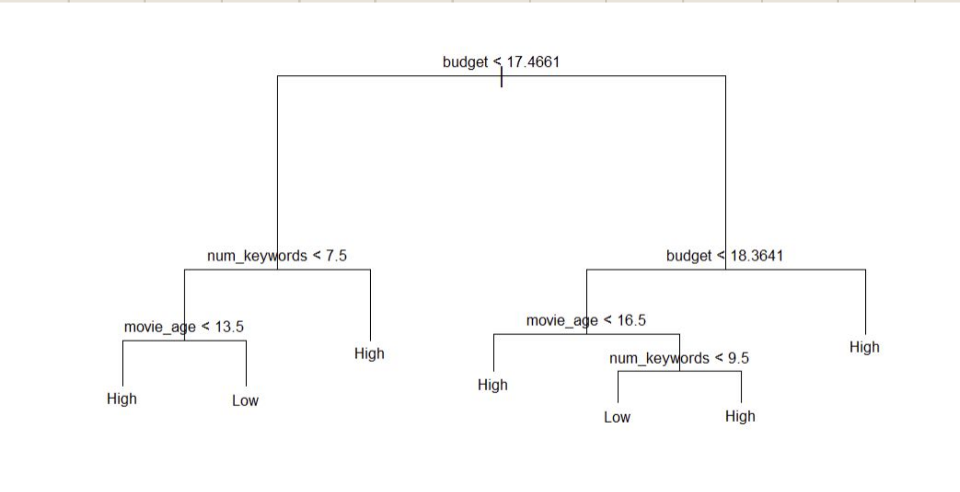




*Figures 2 and 3 - Pre-tuned Classification Tree*

# Finding the Best Tree

- Pruned tree uses three predictors to split on
- Splits on budget, number of keywords, and movie age
- Appears that movies that exceed a budget of $17.5 million are more likely to be popular
- Appears that high budget old movies are popular while low budget old moves are less popular




*Figures 5 and 6 - Tuned Classification Tree Results*

# Creating Random Forest

- Creating Random Forest on same training data for improved accuracy
- Used **ntree = 500 for best** results
- Received **29%** error rate for classification

- Correctly predicts 1,832/2,583 in test set

- Shows most of the variance is explained in budget, num_genres, num_companies, num_langauges, num_keywords, movie_age

```
Call:
 randomForest(formula = popularity_cat ~ ., data = train_tree,      ntree = 500)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 2

        OOB estimate of  error rate: 29.07%
Confusion matrix:
     High Low class.error
High  884 396   0.3093750
Low   355 948   0.2724482
              MeanDecreaseGini
budget               381.08799
num_genres            94.46911
num_companies        139.21070
num_languages         66.52939
num_keywords         269.33790
movie_age            284.78845
[1] "accuarcy"
[1] 0.7260062
[1] "error"
[1] 0.2739938
```

*Figures 7 - Random Forest Results*

# Popularity Conclusion

- Using Classification Tree methods show that popularity is best split and explained by movie budget, age of movie, and number of keywords(themes) a movie encompasses

- Random forest show that number of companies, genres, and languages is meaningful

# Final Conclusions

Based on both models used, the characteristics of movies allow for a strong prediction of both movie revenue and popularity. These methods proved that predictive modeling can be useful in understanding overall movie performance and what factors producers and studios should focus on.

We can conclude that a new movie's success is highly based on budget for the film and that people enjoy newer movies that have a high budget compared to older movies

People seem to like movies that are more complex and have different layers to them.

Future works might include using a test set of future move titles to see how model accuracy is affected and analyze if future movie's success is still based off similar variables.

# Bibliography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- "Driving Local Economies." *Motion Picture Association*, 10 Feb. 2026, www.motionpictures.org/advocacy/driving-local-economies/.
- "Popularity & Trending." *The Movie Database (TMDB)*, developer.themoviedb.org/docs/popularity-and-trending. Accessed 4 May 2026.

# Thank You